LAZARD LEGACY MATH ACADEMY

Financial Literacy Math

Real-World Money Skills for All Ages

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Budgeting & Personal Finance

The 50/30/20 Rule

50% of take-home pay → Needs (rent, food, utilities, transportation)

30% of take-home pay → Wants (entertainment, dining out, shopping)

20% of take-home pay → Savings & debt payoff

Example: Monthly take-home = $3,000. Needs=$1,500. Wants=$900. Savings=$600.

Worked Examples

1. Monthly income $2,500. Apply the 50/30/20 rule. Needs=$1,250, Wants=$750, Savings=$500

2. Groceries $350, rent $800, electric $120, internet $60. Total needs=$1,330. Is this under 50% of $3,000? 50%=$1,500. Yes.

3. You want to save $5,000 in a year. How much per month? $5,000÷12=$416.67/month

4. Annual income $48,000. Monthly take-home (after 20% taxes)? $48,000×0.80=$38,400÷12=$3,200/month

5. Fixed expenses: rent $950, car $300, insurance $150. Variable: groceries $250, gas $80. Total monthly? $1,730

15 Practice Problems

1) Take-home $4,000/month. Apply 50/30/20 rule. 2) Budget $2,200/month. Rent $800, food $300, car $250. How much left? 3) Save $10,000 in 18 months. Monthly amount? 4) Annual gross $60,000, 25% tax. Monthly net? 5) Weekly pay $750. Monthly estimate (×4)? 6) Income $3,500. Needs total $1,900. What % is that? 7) Emergency fund goal: 3 months of $2,800 expenses. Total needed? 8) Reduce spending by 15% from $2,400/month. New amount? 9) Earn $15/hr, 40 hrs/week, 4 weeks/month. Monthly gross? 10) Monthly income $5,000. If needs=$2,200, wants=$1,300, savings=?

Simple & Compound Interest

Formulas

Simple Interest: $I = P \times r \times t$ where P=principal, r=rate (decimal), t=time (years)

Total amount: $A = P + I = P(1 + rt)$

Compound Interest: $A = P(1 + r/n)^{(nt)}$ where n=times compounded per year

Worked Examples

1. \$2,000 at 5% simple interest for 3 years. $I=2000\times0.05\times3=\$300$. A=\$2,300
2. \$5,000 at 4% compounded annually for 2 years. $A=5000(1.04)^2=5000\times1.0816=\$5{,}408$
3. Credit card \$1,500 at 18% annual, paid in 1 year. $I=1500\times0.18=\$270$
4. \$1,000 at 6% compounded monthly for 1 year. $A=1000(1+0.06/12)^{12}=1000(1.005)^{12}\approx\$1{,}061.68$
5. How long to double \$3,000 at 6% simple interest? $3000=3000\times0.06\times t \rightarrow t\approx16.7$ years

15 Practice Problems

1) \$1,000 at 8% simple for 2 yrs 2) \$3,500 at 5% simple for 4 yrs 3) \$2,000 at 6% compounded annually for 3 yrs 4) \$10,000 at 3% simple for 5 yrs 5) Credit card: \$800 at 24% for 1 yr — interest owed? 6) \$500 at 10% simple for 6 months 7) \$4,000 at 7% compounded annually for 2 yrs 8) Which earns more: \$1,000 at 5% simple for 3 yrs, or 5% compound annually for 3 yrs? 9) \$2,500 at 4% simple — how long to earn \$400 interest? 10) \$6,000 at 5% compound, 2 yrs

Taxes & Paycheck Math

Understanding Your Paycheck

Gross pay = total earned before deductions

Net pay = take-home after taxes and deductions

Common deductions: Federal tax, State tax, Social Security (6.2%), Medicare (1.45%), health insurance, 401k

Example: Gross $3,500. Federal 15%=$525. State 5%=$175. SS=$217. Medicare=$50.75. Net≈$2,532.25

Tax Brackets (2024 Single Filer — Simplified)

10%: $0–$11,600 | 12%: $11,601–$47,150 | 22%: $47,151–$100,525

Example: Taxable income $40,000. Tax = 10% on first $11,600 ($1,160) + 12% on remaining $28,400 ($3,408) = $4,568

15 Practice Problems

1) Gross $52,000. Net after 20% total deductions? 2) Hourly $18, 40 hrs/wk. Weekly gross? 3) Annual gross $75,000. Monthly gross? 4) SS deduction on $4,000 gross (6.2%)? 5) Net pay $2,800, total deductions $700. Gross? 6) Gross $45,000. Estimate federal tax (12% bracket)? 7) Earn $22/hr, 45 hrs (5 overtime at 1.5×). Gross for week? 8) Monthly gross $5,200. Annual gross? 9) State tax 4.5% on $3,800 monthly gross? 10) Gross $60,000. SS + Medicare = 7.65%. Annual deduction?

Investing Basics

Growing Your Money

Investing means putting money to work to earn more over time.

Rule of 72: Years to double = 72 ÷ interest rate

Example: At 8% return, money doubles in 72÷8=9 years

$1,000 invested at 7% for 30 years (compounded) = $1,000 × $(1.07)^{30}$ ≈ $7,612

Types of Investments:

• Savings account (0.5–2%) • CDs (2–5%) • Bonds (3–5%) • Index funds/stocks (7–10% average) • Real estate

401(k): Employer-sponsored retirement account. Contribute pre-tax dollars. Many employers match contributions.

10 Practice Problems

1) Rule of 72: how long to double at 6%? 2) $2,000 at 8% for 10 yrs (use $A=P(1+r)^t$) 3) $5,000 at 7% for 20 yrs 4) How much to invest now to have $10,000 in 5 yrs at 5%? 5) Employer matches 3% of salary $50,000. Annual match? 6) $500/month in index fund at 8% for 30 yrs ≈ what total? 7) Rule of 72: time to double at 9%? 8) $10,000 at 6% for 15 yrs 9) $1,200/yr in 401k, employer matches 50%. Annual total contribution? 10) Start investing $200/month at 25 vs $400/month at 45. Which likely yields more by 65?

Credit Scores & Debt Payoff

Understanding Credit

Credit score range: 300–850. 720+ = good. 800+ = excellent.

What affects your score:

• Payment history (35%) • Amounts owed (30%) • Length of credit history (15%) • New credit (10%) • Credit mix (10%)

Debt Payoff Methods

Avalanche method: Pay highest interest rate first (saves most money)

Snowball method: Pay smallest balance first (builds momentum)

Example: $5,000 at 20% APR, minimum payment $100/month. Interest per month = $5,000×0.20/12=$83.33. Only $16.67 goes to principal!

10 Practice Problems

1) Credit card $3,000 at 22% APR. Monthly interest? 2) Pay $200/month on $3,000 balance at 0% interest. Payoff time? 3) Which debt to pay first (avalanche): Card A $500 at 25%, Card B $2,000 at 15%? 4) $8,000 student loan at 5%. Monthly interest? 5) Monthly income $3,500. Total debt payments $800. Debt-to-income ratio? 6) 3 debts: $200, $500, $1,200. Snowball order? 7) Score improves 40 points, now 665. Original score? 8) Minimum payment $50 on $1,000 at 18% APR. Monthly interest? 9) Pay off $6,000 in 24 months. Monthly payment? 10) Two credit cards: $1,500 at 18%, $900 at 24%. Which to pay first (avalanche)?

30 Real-World Financial Word Problems

Answer Key Included

1. Marcus earns $52,000/year. After 22% in taxes, what is his monthly take-home? Answer: $\$52{,}000 \times 0.78 = \$40{,}560$/yr ÷ 12=$3,380/month

2. Tanya saves $250/month. How long to save $6,000? Answer: $6,000÷$250=24 months

3. Car costs $18,000 at 6% simple interest for 4 years. Total interest? Answer: $\$18{,}000 \times 0.06 \times 4 = \$4{,}320$

4. Gregory has 3 credit cards: $800 at 19%, $1,200 at 23%, $500 at 15%. Using avalanche, which first? Answer: $1,200 at 23%

5. Monthly budget: income $3,200, rent $900, car $280, insurance $120, food $350, utilities $150. Remaining? Answer: $3,200–$1,800=$1,400

6–30. [Complete the remaining 25 real-world financial problems covering budgeting, interest, taxes, investing, and credit management. Review all answers with an instructor.]